

PAPER CASE 03 / NATURE 2024

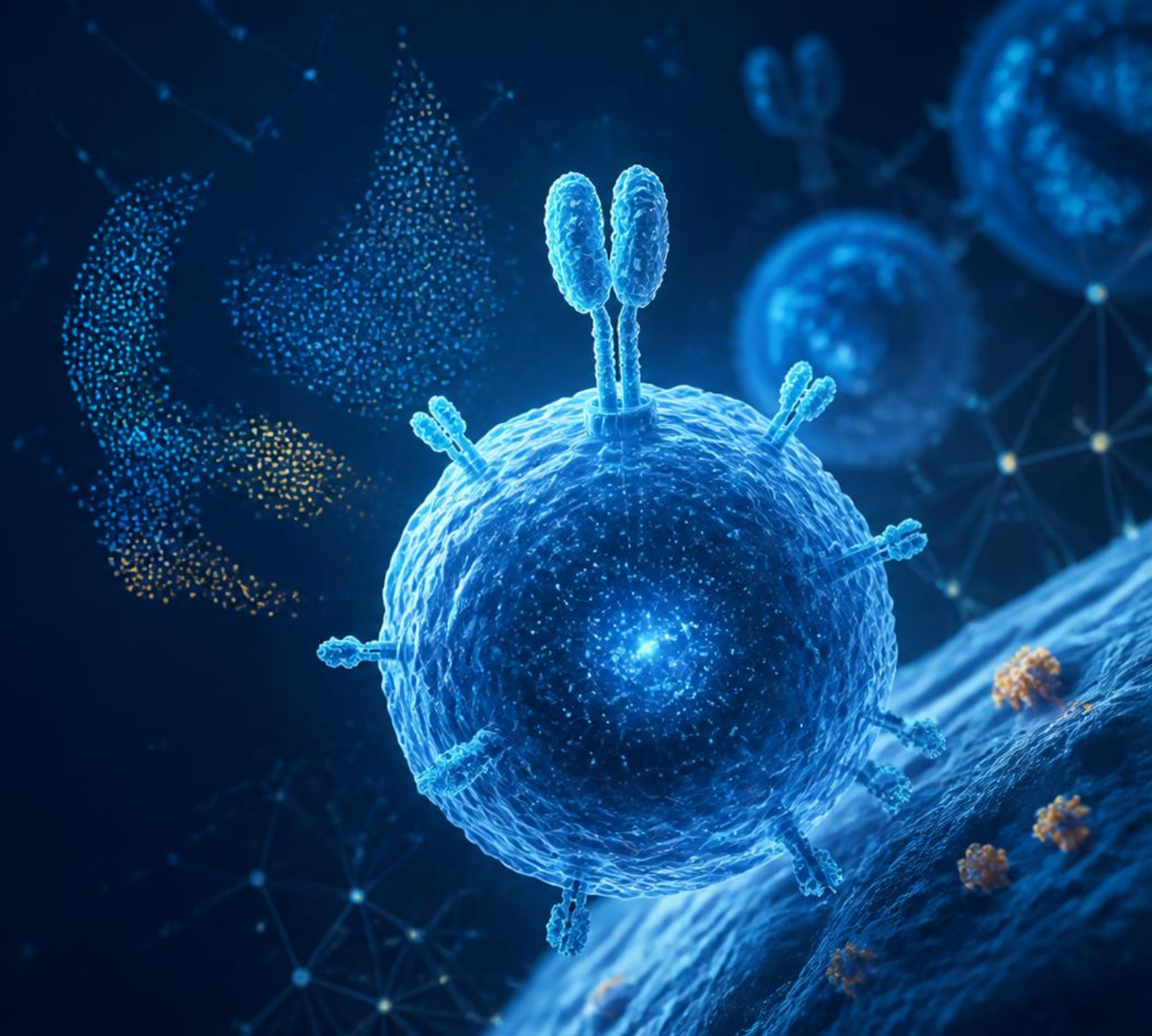
When does a CAR T cell stay fit?

A single-cell atlas of an 8-year remission.

Bai et al. · Nature 634 · 702–711 · 2024
82 patients · 695,819 analyzed cells
CONCEPTUAL COVER VISUAL

CLINICAL → CELLULAR → TESTED

DOI 10.1038/s41586-024-07762-w



The phenotype stays visible for years

The paper begins with a durable clinical readout: B-cell aplasia after CAR T infusion.



BCA = B-cell aplasia • BCA-L = longest-remission stratum

8.4 years
median BCA-L duration

This is not a single endpoint.
It is a clinical trace that motivates the atlas.

OBSERVED

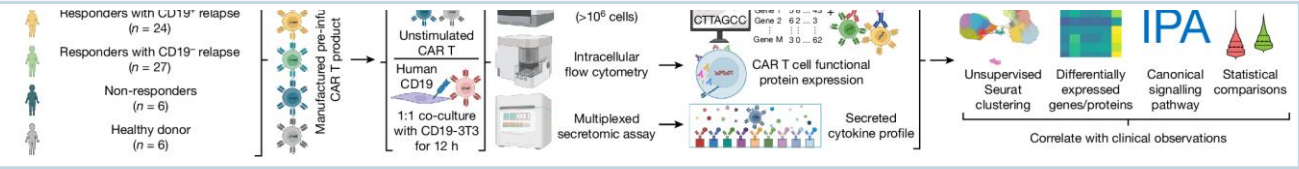
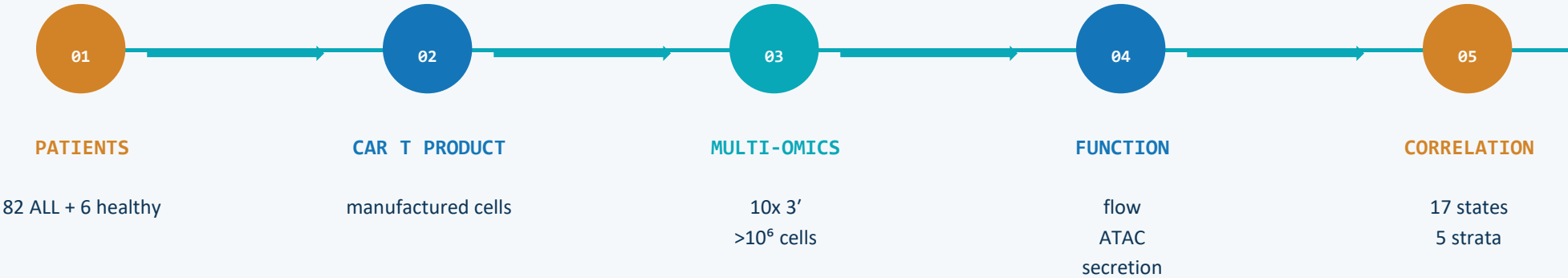
long B-cell aplasia



what sustains it?

The study turns persistence into a measurable system

Five linked layers connect patients, products, cell states, function, and clinical correlation.



Source: Fig. 1a · editorial redraw of study pipeline

695,819
analyzed cells

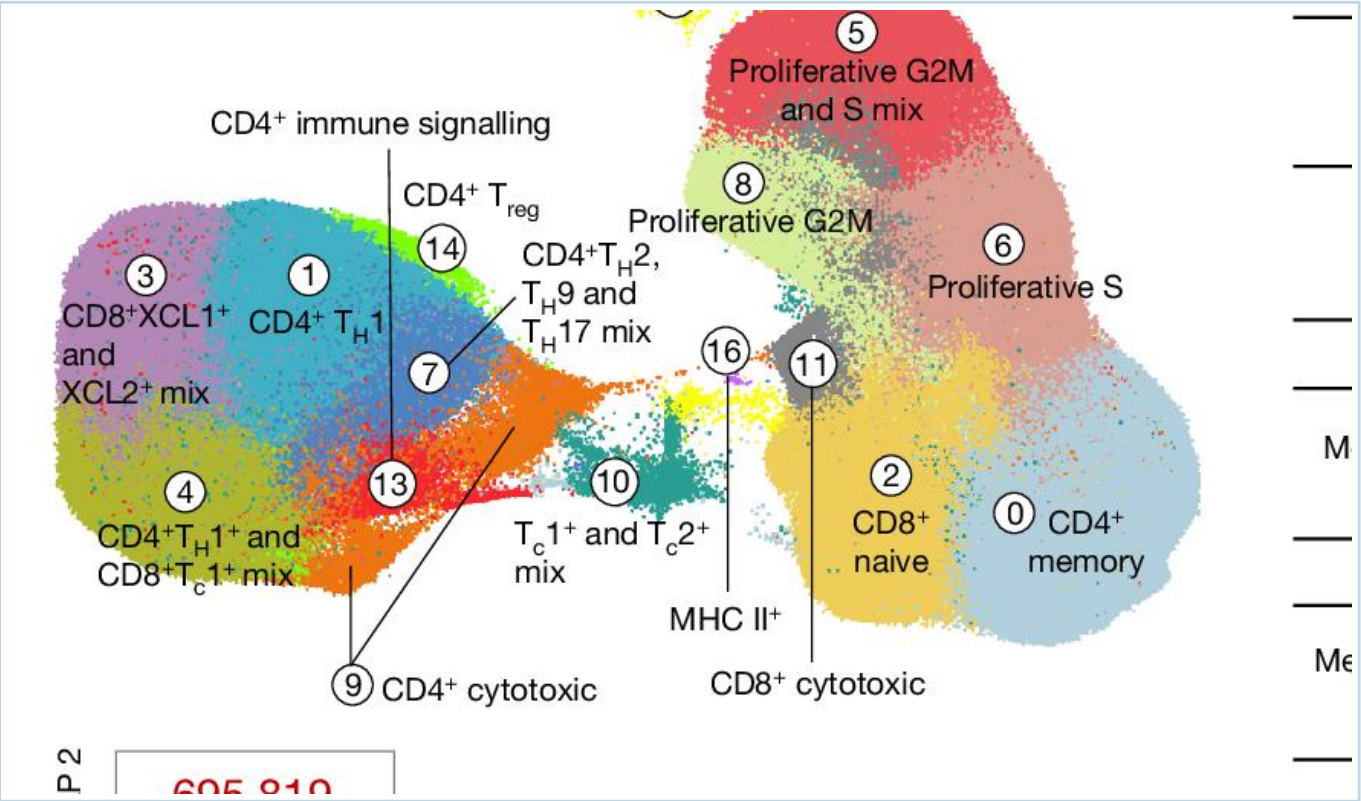
17
cell states

READOUT LAYERS

transcriptome · flow cytometry · secretome · ATAC

A field of 695,819 cells resolves into 17 states

This map is the paper’s transition from clinical phenotype to cellular state.



FOUR STATE FAMILIES

- NAIVE**
memory / early state
- CYTOTOXIC**
effector program
- REGULATORY**
signalling program
- PROLIFERATIVE**
expansion program

WHAT THE MAP ENABLES

state composition → function → clinical strata

Source: Fig. 1b • UMAP and cluster annotation

Original cluster colors are preserved inside the source figure; family labels are editorial grouping, not new clusters.

17
cell states

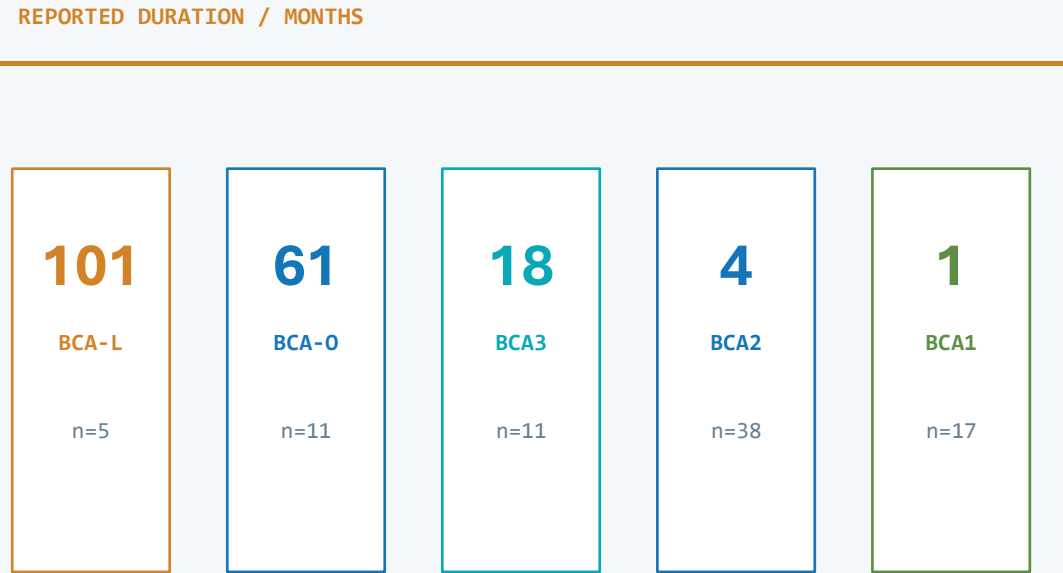
695,819
cells

BCA strata create the clinical gradient

The groups are ordered by reported duration; the slide does not imply a continuous fitted curve.

cs	BCA-L (n = 5)	BCA-O (n = 11)	BCA3 (n = 11)	BCA2 (n = 38)	BCA1 (n = 17)
ort	5	NA	5	21	11
ort	NA	11	6	17	6
age	14.5 ± 1.4	13.3 ± 2.0	13.3 ± 1.9	9.7 ± 1.0	10.4 ± 1.2
e)	(9.3–16.2)	(6.0–29.1)	(5.0–24.5)	(1.7–27.2)	(3.3–21.5)
ad	Yes	Yes	Yes	Yes	Partial (11/17)
duration	101 ± 5.0	61 ± 1.8	18 ± 2.9	4 ± 0.4	1 ± 0.2
e)	(82–106)	(48–72)	(12–43)	(3–11)	(0–2)
ved	No	No	Yes	Yes	Yes
to relapse	NA	NA	18.5 ± 3.3	8.5 ± 2.2	2.1 ± 1.1
e)	NA	NA	(12.0–45.6)	(2.5–60.2)	(1.0–9.3)
..	101 ± 5.0	61 ± 1.8	18 ± 2.9	4 ± 0.4	1 ± 0.2

Source: Fig. 1c - demographics and follow-up

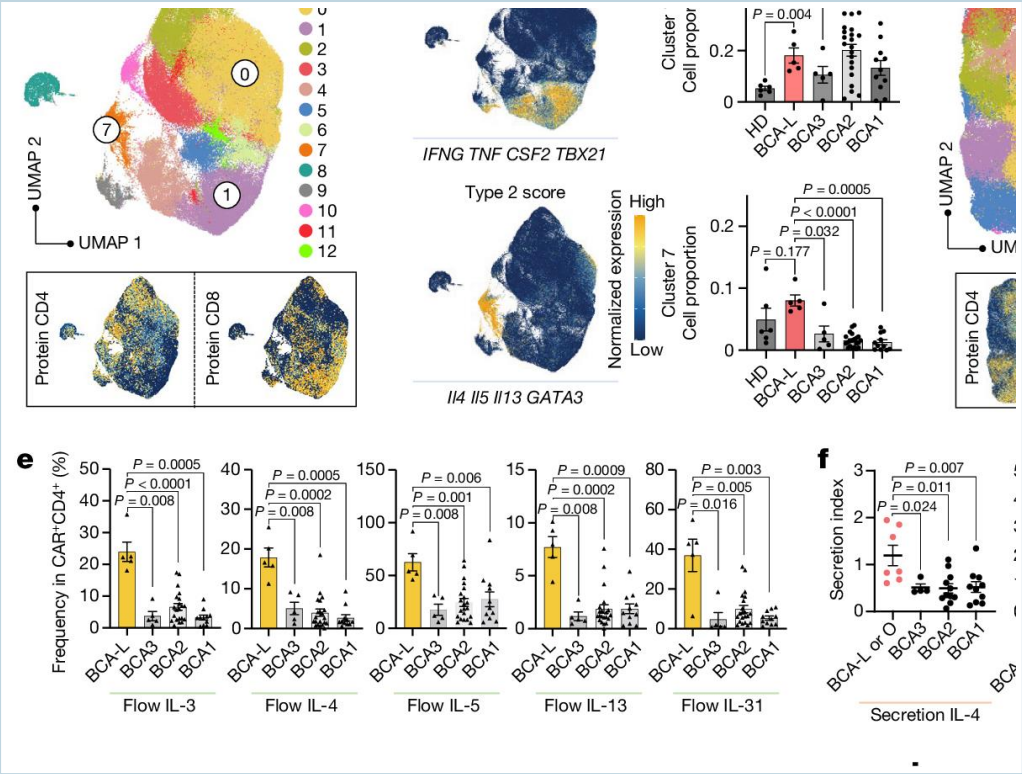


READING RULE

BCA-L is the extreme anchor; later pages ask what distinguishes that state.

Long-remission cells show a type 2-like program

Three molecular readouts converge on the same state; the title stays associative.



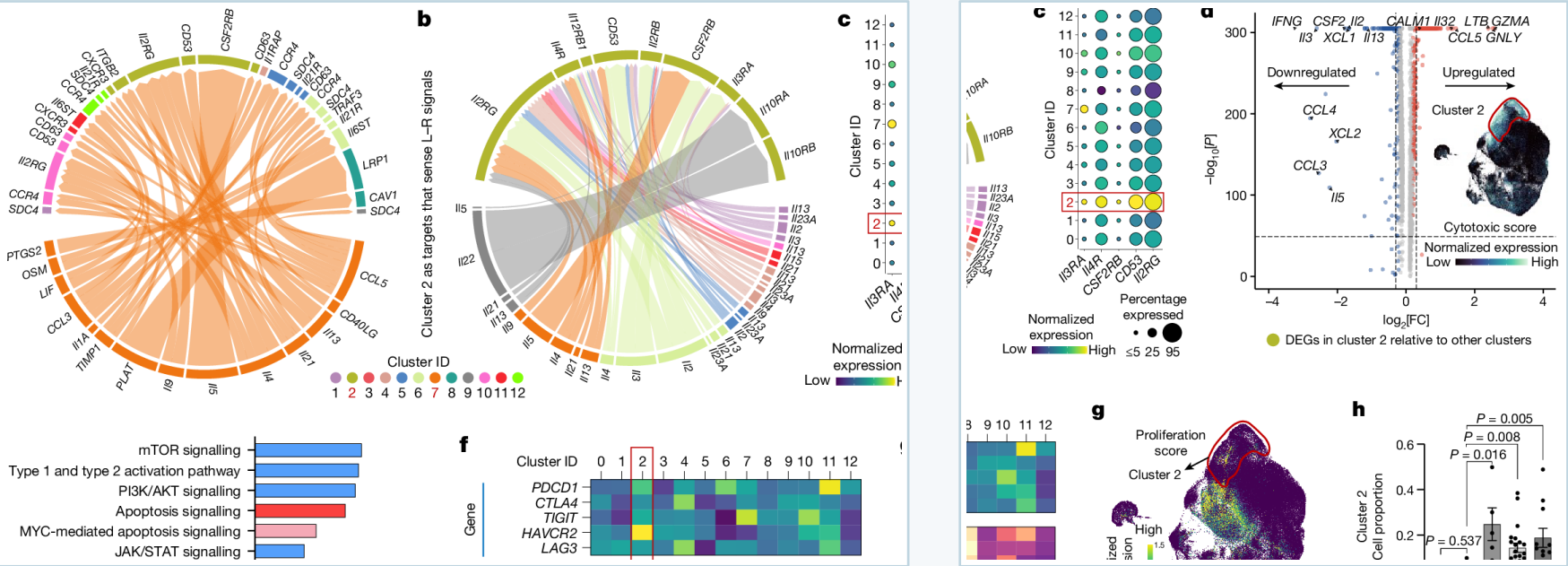
Source: Fig. 2a-d · type 2 score and validation

one coherent type 2 state

Functional context: repeat stimulation → tumour-cell lysis

Cluster 2 emerges as a candidate regulatory hub

The paper proposes a network relationship; this page does not turn it into complete causal proof.



Source: Fig. 3a-b · ligand-receptor network

Source: Fig. 3c-d · cluster 2 and DEGs

13.9%
cluster 2

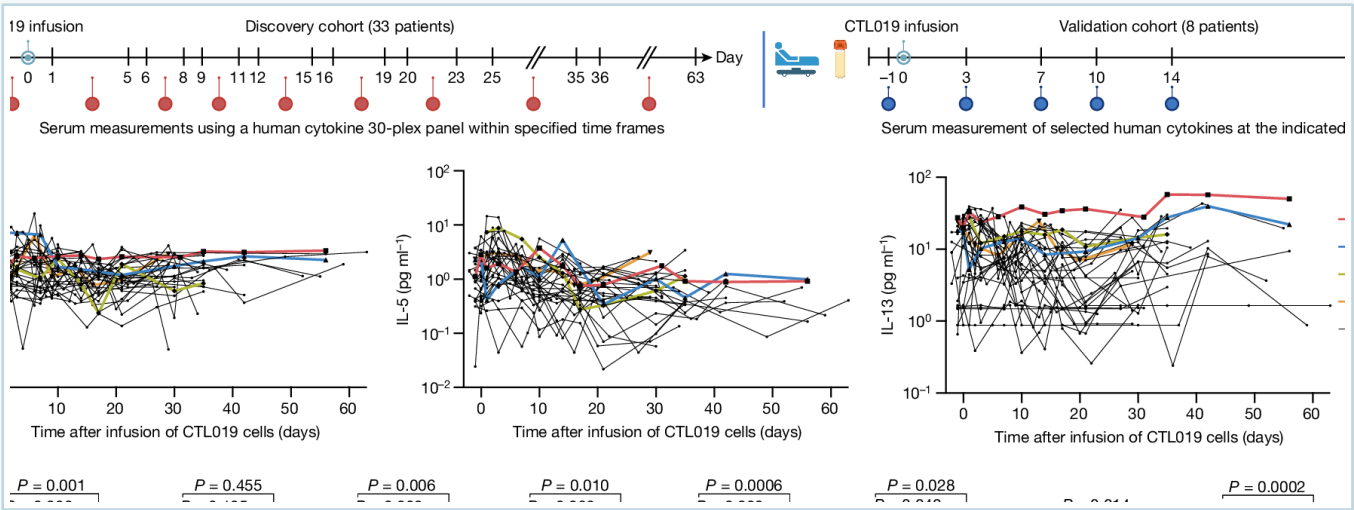
BOUNDARY

candidate network → hypothesis for dysfunctional-subpopulation regulation

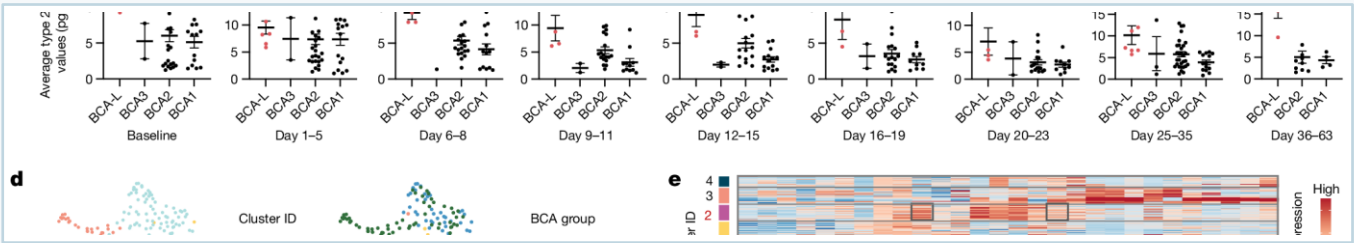
Analysis layers: ligand-receptor links · mTOR signalling · cluster-2 differential expression

The type 2 signal survives time, not just a snapshot

Serial serum measurements extend the cellular observation into a longitudinal readout.



Source: Fig. 4a-b · collection timeline and time series



Source: Fig. 4c-e · cytokine windows and heatmap

Source: Fig. 4. Longitudinal serum evidence supports persistence of the biology; it does not replace the cell-state analysis.

345
measurements

30
cytokines

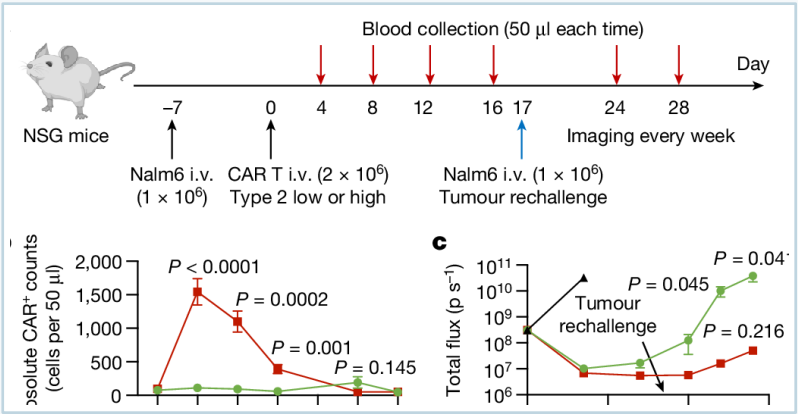
33
patients

TYPE 2 WINDOW

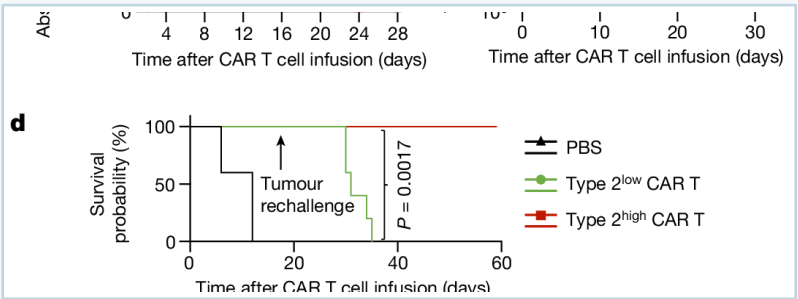
IL-4 · IL-5 · IL-13
days 1–63
discovery 33 · validation 8

The state is tested as function in a recall model

The mouse experiment tests expansion, tumour control, and response after rechallenge.

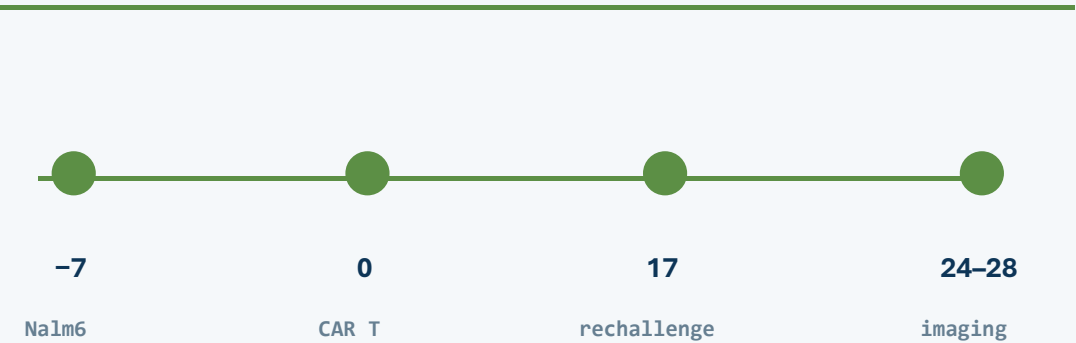


Source: Fig. 5b-c · expansion and tumour flux



Source: Fig. 5d · survival after rechallenge

EXPERIMENTAL SEQUENCE



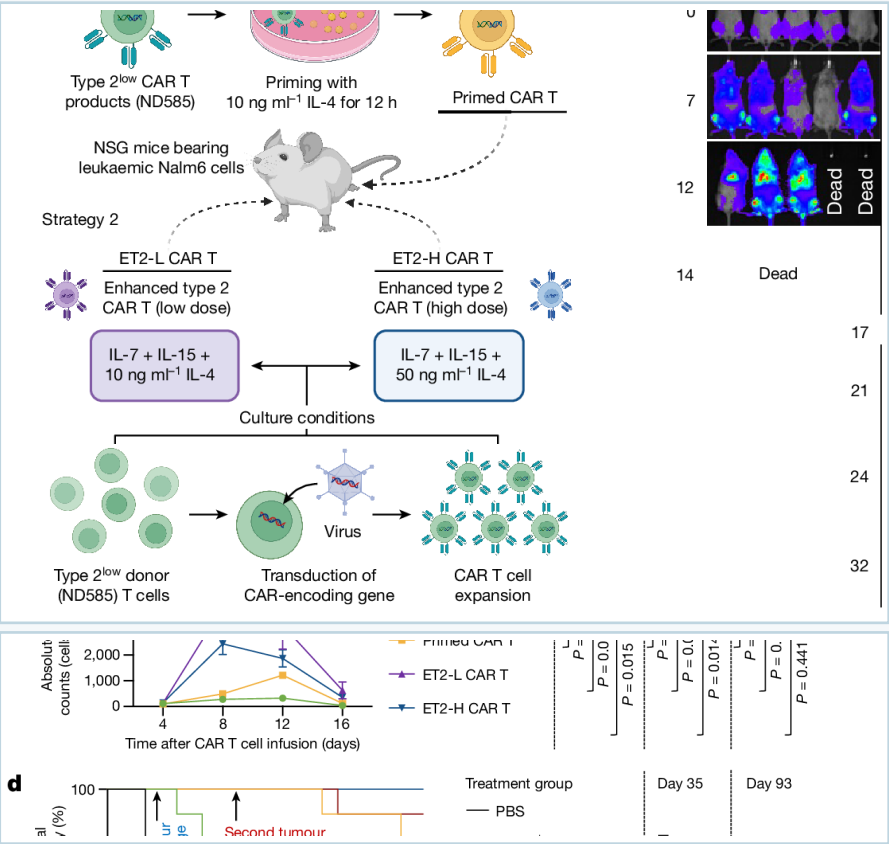
READOUT

Type 2-high cells expand more and show stronger recall in the reported model.
Selected readouts: n=5 mice.

Comparison: type 2-low versus type 2-high CAR T cells

IL-4 provides a preclinical manufacturing direction

The final figure asks whether a type 2-like program can be introduced or primed experimentally.



Source: Fig. 6c-d · expansion and survival

EXPOSURE → PRODUCT → TEST



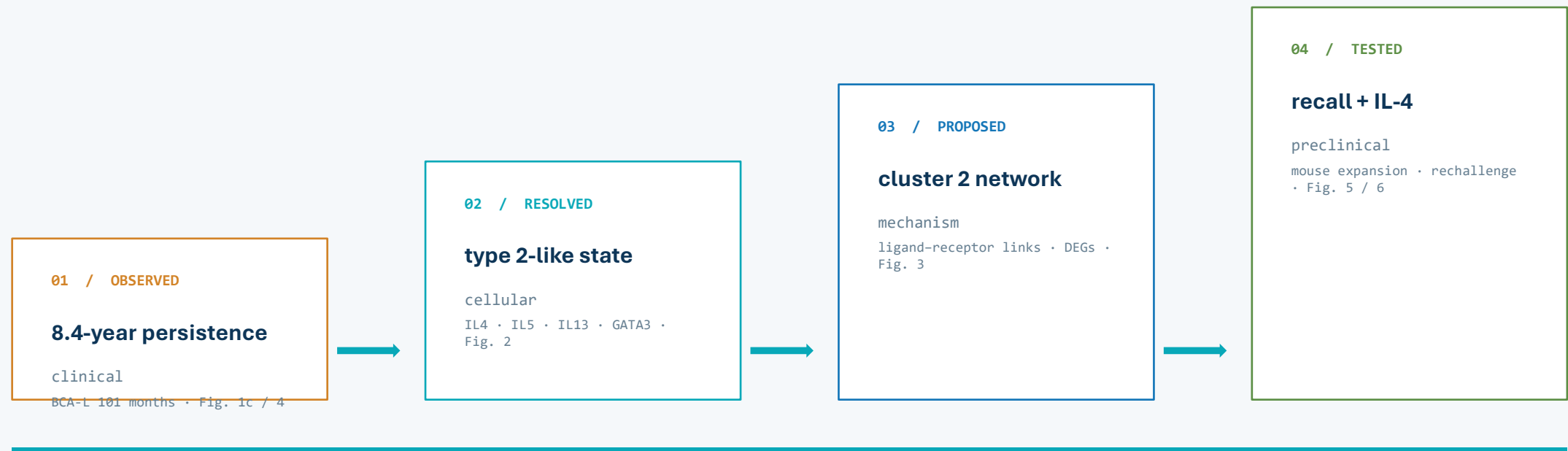
Culture context: IL-7 + IL-15; dose labels remain paper-specific

BOUNDARY

promising lever, still preclinical

The argument is strongest when evidence types stay separate

A disciplined reading keeps association, state definition, mechanism, and intervention distinct.



association → state definition → candidate explanation → functional test / engineering lever

Type 2 function may sustain fitness.

A durable clinical phenotype becomes a recognizable cell state, a candidate circuit, and a preclinical lever.

Contribution

connects persistence to cell state and function

EVIDENCE MAP

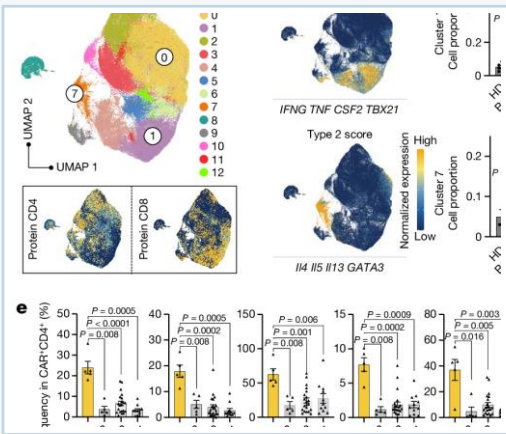


Fig. 2 • cellular state

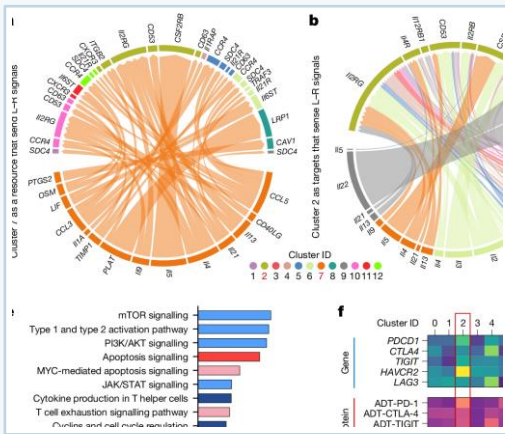


Fig. 3 • candidate network

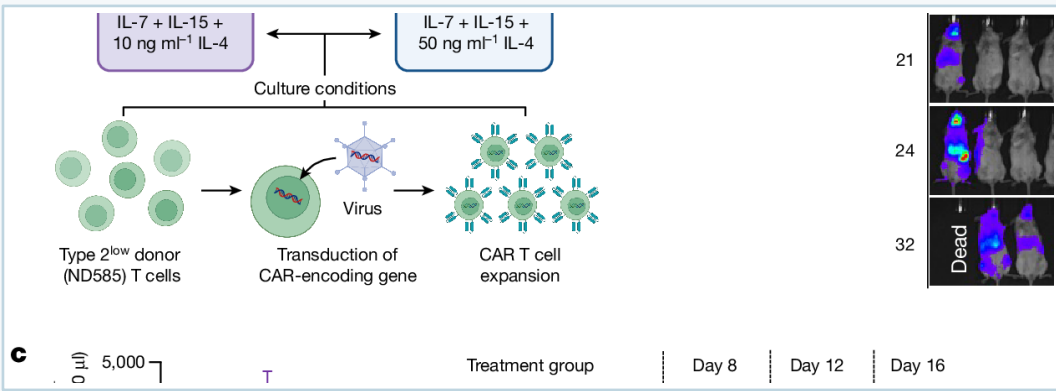


Fig. 6 • engineering direction

LIMITS

Optimal clinical type-2 level remains to be established; preclinical models do not establish efficacy.